

Learner preferences and prior knowledge in learner-controlled computer-based instruction

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Abstract This study examined the effects of prior knowledge, learner preference for control, and type of control (learner or program) on the achievement of middle-school students in a computer-based instructional program on adding and subtracting integers. Students were blocked by preference-for-control scores and randomly assigned to either a learner-control or program-control version of instruction. A significant three-way interaction (prior knowledge $\times$ preference scores $\times$ type of control) revealed that students with high prior knowledge achieved better on the posttest when their preference for control was matched with the type of control they received, whereas students with low prior knowledge achieved better when their preference was mismatched. A significant three-way interaction on attitude reflected the same pattern found in the interaction for achievement scores. The overall results indicate that matching learner preference to the type of program they receive is an effective strategy for high-prior-knowledge students but not for those with low prior knowledge.

Keywords Learner control · Learner preferences · Prior knowledge · Computer-based instruction · Mathematics

The benefits promised by giving learners control over their instructional decisions are alluring to many educational technologists. Researchers have postulated that giving learners control over their instruction may lead to increased motivation, improved achievement and performance, and more positive attitudes about learning (Kinzie 1990; Swaak and de Jong 2001; Triantafillou et al. 2003; van Gog et al. 2005). Based on the work of Merrill (1975, 1980, 1984), Kinzie (1990) envisioned learners who, when given control over their learning, would be both intrinsically motivated to learn and capable of

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making sound instructional decisions. These learners would exhibit qualities such as taking the initiative to pursue new knowledge and successfully making choices about what to learn and how best to learn it. Van Gog et al. (2005) similarly suggested that learners who were allowed to control elements of their instruction were more likely to develop the skills needed to make decisions about their learning and exhibit higher levels of motivation towards learning.

The results of several studies on learner control in computer-based instruction (CBI) have indicated that giving learners control over elements of instruction may lead to improvements in achievement (Corbalan et al. 2006; Kinzie et al. 1988; Shyu and Brown 1992; van Merriënboer et al. 2002). Van Merriënboer et al. (2002) gave communication science students learning computer-programming skills control over the type of problem format they received during instruction, one with limited guidance and one with no guidance. When compared to a group that received only one type of problem format (no guidance), the students with control performed significantly better on a test of transfer. Shyu and Brown (1992) found that university undergraduates who were allowed to pause, rewind, and skip instruction while learning a procedural task from an interactive videodisc had significantly higher task performance levels than their counterparts who were not given such control.

The positive outcomes of some learner-control studies have prompted instructional designers to consider the issue of learner control when designing instruction using the Internet or hypermedia (Chen and Macredie 2002; El-Tigi and Branch 1997; Gerjets and Scheiter 2003; Schmidt and Ford 2003). El-Tigi and Branch (1997) noted that learner control is an inherent feature of learning on the World Wide Web and suggested strategies for maximizing learning on the Web. Gerjets and Scheiter (2003) recommended that cognitive load theory include cognitive factors such as learner goals and processing strategies to successfully apply it to the design of learner-controlled learning environments.

Some cognitive load theorists (Bannert 2002; Salden et al. 2006; van Gog et al. 2005) see learner control as a way to potentially reduce cognitive load and improve the learning outcomes in computer-based instruction. These theorists suggest that giving learners control over aspects of their instruction allows them to better negotiate the cognitive demands placed on them. Bannert (2002) suggested that this might be the case for the learner-control subjects in the van Merriënboer et al. (2002) study who performed better on a test of transfer than the subjects given no control. Van Gog et al. (2005) hypothesized that gradually giving learners more control over instruction as their skill level increased would not only improve performance during training situations, but would help foster the learner's ability to diagnose and correct problems they encounter in the future.

Giving learners control over instructional elements does not necessarily lead to improvements in learning in computer-based instruction. Schnackenberg and Sullivan (2000) found that preservice teachers who could control the amount of practice received when learning to write objectives performed no better than those who did not have such control. A study by Eom and Reiser (2000) reported that sixth- and seventh-grade students who were allowed to control their instruction performed significantly worse than those who did not have control. Several researchers (Farrell and Moore 2000–2001; Morrison et al. 1992; Swaak and de Jong 2001; Yang and Chin 1996) have reported that giving learners control over their instruction results in little or no improvement in achievement.

The nature of the control provided to learners may be an important factor in determining its effectiveness. If the learner-control feature enables learners to bypass some of the instruction, one might expect learners who do so to perform less well than those who do not have that option. If, on the other hand, the learner-control feature enables learners to add

instruction, one might expect those who do so to perform better than those who do not have that option. Some researchers have studied the effects of learner control in versions of the same program that varied in length and allowed subjects in the longer (or full) version to bypass part of the instruction and those in the shorter (lean) version to add instruction to their basic program (Carrier et al. 1985; Freitag and Sullivan 1995; Hannafin and Sullivan 1996). In the full version, participants received more instances of selected elements of instruction (e.g., more practice, more examples, more review), whereas in the lean version they received fewer instances of these instructional elements. In each study, the full version of the instruction contained the same amount of instruction as a learner-controlled lean version in which the learner exercised the option to select all of the optional elements. Conversely, the lean version was the same length as a learner-controlled full version in which the learner selected the option to bypass all of the available optional elements.

Using a lean and full version of instruction allows learner control to be examined from two perspectives within the same study. Participants under learner control receive either a lean program in which they have the control to add more instruction or a full program in which they have the control to bypass some of the instruction. Hannafin and Sullivan (1995) found that 9th-grade and 10th-grade students learning geometry performed significantly better under learner control than under program control. However, learner control was not found to be superior to program control by Carrier et al. (1985) or by Schnackenberg and Sullivan (2000). Schnackenberg and Sullivan (2000) attributed this to the amount of practice received by each group. Participants in the learner-control groups who could choose to add or bypass practice screens viewed nearly the same average number of practice screens as the program-control groups.

Individual learner preferences for the instruction they receive may also account for the success of some learners who are given control over their instruction (Bannert 2002; Freitag and Sullivan 1995; van Merriënboer et al. 2002). Van Merriënboer et al. (2002) suggested that subjects under learner control performed better because the instruction was better able to meet their preferred style of learning. Freitag and Sullivan (1995) found that adult participants working at a major corporation performed significantly better when their preference for the amount of instruction they received (high or low) matched the version of instruction they received (full or lean). However, Schnackenberg et al. (1998) found that preservice teachers learning to design instruction did not perform better when their preference for the amount of instruction (high or low) matched the version of instruction they received (full or lean). These subjects achieved significantly more in the full version of the instructional program even when they indicated a preference for less instruction. Other studies have demonstrated that learners often prefer a version of instruction that is easier to understand or quicker to complete instead of the one that helps them achieve more (Farrell and Moore 2000–2001; Gray 1987).

Another factor that may help determine the success of learners in a learner-control condition is their prior knowledge of the subject-matter content (Klein and Keller 1990; Merrill 2002; van Gog et al. 2005). Klein and Keller (1990) studied ability and learner control using the instructional program designed by Carrier et al. (1984) on advertising concepts. They found that scores on a test of mental ability accounted for more than 40% of the variance on the increase in achievement from pretest to posttest. Reisslein et al. (2007) found that engineering students with high prior knowledge of basic electrical circuits performed better on retention tasks under a fast rate of transitioning to independent problem solving, and that those with low prior knowledge performed better under a slow rate of transitioning. This is consistent with the reports of researchers who have suggested

that learner control would be a greater benefit to learners with higher levels of prior knowledge or ability (Corbalan et al. 2006; Lawless and Brown 1997; Merrill 2002).

Other researchers have found that learners with a high level of prior knowledge of subject-matter have performed better in learner-control conditions than those with a lower level of prior knowledge (Gay 1986; Shin et al. 1994; Williams 1996). Gay (1986) found a significant interaction between prior knowledge and control type with undergraduates in an introductory Biology course. Participants with low prior knowledge had significantly lower posttest scores under learner control than those with low prior knowledge under program control and those with high prior knowledge under either learner or program control. Shin et al. (1994) also found that second-grade students with low prior knowledge performed better when given less control over a hypertext instructional program on the four food groups. It seems likely that learners with lower prior knowledge find it difficult to learn new information while at the same time making choices about adding or bypassing elements of their instruction. Their lower levels of prior knowledge may also affect their potential to be aware of what they do not know and to make choices that will help them learn it.

The topic of learner control continues to be an important one in the field of educational technology. Many designers of web-based and multimedia-based learning environments are concerned over how such control should be given to learners without confusing them or otherwise interfering with their learning (Alomyan 2004; Chen and Macredie 2002; Lawless and Brown 1997; Wang and Sutton 2002). Researchers interested in cognitive load theory consider it important to study the effects of giving learners control over their instruction (Gerjets and Scheiter 2003; van Gog et al. 2005) and also with designing learner-controlled instruction that meets individual learner preferences (Bannert 2002; van Merriënboer et al. 2002). While prior knowledge and preference for amount of control have been studied separately in past learner-control research, the authors could not find any research in which they were examined in the same study.

This study was conducted to examine the effects of learner control and program control in computer-based instruction on the mathematics achievement of students with differing amounts of prior knowledge and differing levels of preference for control. The primary research questions investigated in this study were as follows:

1. Do learners achieve better in a full version of a computer-based instructional program in which they have no control over the amount of instruction (program control) or in a full version in which they have control over the number of examples and amount of practice they receive (learner control)?
2. Do learners achieve better in a computer-based program when their preference for amount of learner control is matched with the amount they receive?
3. Are there any statistical interactions between learners' prior knowledge on the instructional task, learner preference for control, and type of control (learner or program) that affect learner achievement?
4. Do amount of preference for control and prior knowledge influence learners' choices of amount of instruction in a computer-based instructional program?

The first three research questions above were also investigated as they relate to learner attitudes as well as to achievement. Data were also collected and analyzed on time-in-program, optional screens viewed under learner control, and participant interviews.

Previous research on learner and program control indicated that learners achieve better in a full instructional program than in a lean one (Hicken et al. 1992; Hannafin and Sullivan 1996; Schnackenberg et al. 1998). Therefore the basic version of the program used in this study was designed as a full one, from which students under learner control had

the option to skip parts of the instruction. However, the existing research evidence related to the effects of prior knowledge and learner preference for control did not seem strong enough to the researchers to warrant directional hypotheses or predictions about the effects of these factors.

Methods

Participants

Participants were 99 sixth-grade and seventh-grade students (44 male and 53 female) from a middle school in the northeastern United States, 70 from sixth grade and 29 from seventh grade. The school has a population of approximately 700 students, about 30% of whom are minority (18% Hispanic, 9% African-American, 3% Asian), and is located in a middle-class to upper-middle class neighborhood. The school contains a computer lab with 24 IBM-compatible computers. Students regularly visit the lab and work with the computers on tasks such as accessing information on the Internet and creating documents using Microsoft Word or other productivity tools. The school was selected for participation in the study because of the students' familiarity with using computers and their ability to navigate websites for instructional purposes. Also, the content of the instructional program was part of the regular mathematics curriculum at the school.

Instructional materials

The computer-based instructional program called *Operations with Integers* was designed by the primary researcher, who has 6 years of experience teaching mathematics at the secondary (grades 7–12) level, to teach students addition and subtraction with two one-digit integers. The program was designed for delivery on the Internet using Macromedia Flash as the authoring tool. Two content and instructional experts, in addition to the researcher, reviewed the program. The program was field tested with a small group of seventh-grade students prior to the study and revised based on the results of the pilot test.

The program contained a total of eight learning objectives divided into two modules with four objectives in each module. Module 1, Addition, and Module 2, Subtraction, each covered the addition and subtraction of: (a) a positive and a positive integer, (b) a positive and a negative integer, (c) a negative and a positive integer, and (d) a negative and a negative integer. Each module took from 25 to 35 min to complete. All participants received the modules in sequential order. Students advanced through the program by using the mouse and selecting a NEXT button to view the next screen. The NEXT button appeared at the bottom-right corner of each instructional screen.

The eight objectives were taught using screens that led the learner through the information, examples, and practice and feedback for each objective. Information screens contained directions for using the program or text that explains the process of adding or subtracting integers. The examples were presented to the learner after the information screens for each objective. Example screens contained problems and explanations on adding and subtracting integers using the number line as an aid. The examples were animated to mimic solving each problem using the number line.

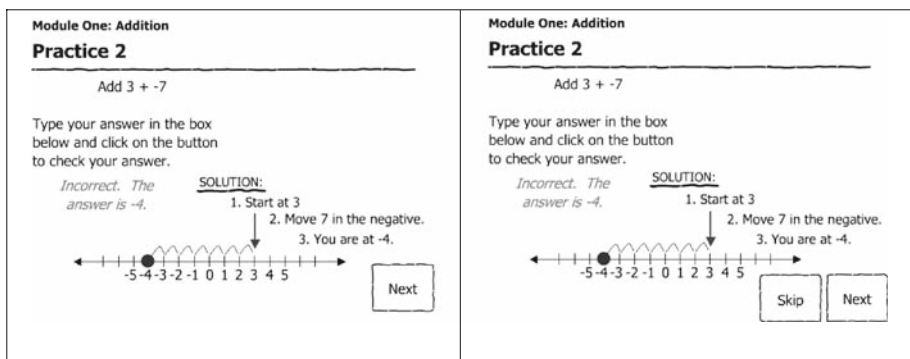
The practice screens contained practice items that allowed the learner to enter information into the computer. The practice items were constructed-response questions for

which students entered a solution on the screen. Students were not allowed to advance to the next practice problem until answering the preceding one. An example of a practice item was “Add $3 + -7$ ”. The feedback for each problem stated that the answer was correct or incorrect. If incorrect, it displayed the correct solution and solution process. A NEXT button was available on each feedback screen so that students could then proceed to the next practice problem.

The computer-based instructional program was developed in two versions, a program-control and a learner-control version. The program-control participants advanced through the program by using the mouse and selecting the NEXT button to view the next screen, viewing all information screens, all examples, and attempting all practice-and-feedback opportunities. The program-control version contained a total of 249 screens, including 54 information screens, 131 example screens (examples ranged in length from one to eight screens) and 64 practice-and-feedback screens (4 practice screens and 4 feedback screens for each of the eight objectives).

The learner-control version contained the same 249 screens as the program-control version, plus an additional three screens near the beginning that explained how to use the SKIP button that allowed students to skip instruction. However, the learner-control version differed from the program-control version by giving participants the option to stop viewing the examples for each objective after both the first and second of three examples for the objective, and to exit the practice for an objective after each practice item for the objective. Participants in the learner-control version advanced through the instruction using the mouse to select the NEXT button, but also had a SKIP button during the example screens and the practice screens when allowed to control the instruction. Figure 1 contains a sample practice screen from both the program-control and learner-control versions of instruction.

Depending upon their choices, learner-control participants could receive from one to three examples and from one to four practice-and-feedback opportunities per objective. In total, the learner-control version offered from 8 (one for each of the eight objectives) to 24 (three per objective) examples for learners to view and from 8 (one constructed-response practice item per objective) to 32 (four such items) practice-and-feedback opportunities. Thus, the total number of optional elements (examples and practice) a learner-control participant could choose to skip was 40 elements, consisting of 16 (of the 24) examples and 24 (of the 32) practice-and-feedback opportunities. The minimum total number of screens a participant could view in the learner-control version was 123: all 54 information screens,



Practice Screen

Fig. 1 Sample practice screen from program-control (left) and learner-control (right) versions

53 example screens from 8 examples, and 16 practice-and-feedback screens from 8 practice-and-feedback opportunities. As for program control, the maximum a learner-control participant could view was all 249 instructional screens plus the three additional explanatory screens.

Criterion measures

Learner Preference Scale

A Learner Preference Scale was designed to assess students' preference for control over instruction. The researchers administered the scale to assess students' preference for control over decisions a teacher usually makes when students are learning new math skills. The researchers created the scale, which consists of eight Likert-type items, because no similar instrument could be found to assess the control preferences of subjects in learner-control studies. The full scale is presented in Table 1.

Each of the eight items was scored on a five-point basis ranging from strongly agree (4) to strongly disagree (0). The maximum score was 32 points, and the mean score was 22.90. Cronbach's coefficient alpha for the scale, based on the students in this study, was .82.

Table 1 Learner Preference Scale

Directions: Please circle your level of agreement with each statement, from strongly agree to strongly disagree.

	Strongly Agree	Agree	Neither Agree nor Disagree	Disagree	Strongly Disagree
When I am being taught something new in math class...					
1. I would prefer to make the decisions the teacher makes about how much instruction I get.	SA	A	N	D	SD
2. I would like it if I could choose to have less instruction than the teacher gives me.	SA	A	N	D	SD
3. I would rather decide how I learn a new math concept instead of having the teacher decide.	SA	A	N	D	SD
4. I know better than the teacher how many problems I should do.	SA	A	N	D	SD
5. I would like to decide, without the teacher's help, how many worked-out example problems I get to see.	SA	A	N	D	SD
6. I would like to decide, without the teacher's help, how many problems I do.	SA	A	N	D	SD
7. I would like to decide when to stop doing problems instead of having the teacher decide.	SA	A	N	D	SD
8. I know better than the teacher how much instruction I need.	SA	A	N	D	SD

Pretest

The students completed a 12-item constructed-response pencil-and-paper pretest 3 weeks before using the instructional program. The researchers intentionally limited the range of integers on the pretest problems from 5 to -5 , with no two-digit correct answers, in an attempt to obtain a broader distribution of scores and to avoid clustering of subjects at low-score levels. The following is an example of an addition item from the pretest: *Add $3 + -2$* . The mean pretest score was 7.37 ($SD = 2.10$) or 61% correct. Cronbach's coefficient alpha for the pretest was .59.

Posttest

The primary measure of achievement was a 24-item (double the number of items on the pretest) constructed-response pencil-and-paper posttest administered at the end of the instruction. The posttest covered all eight objectives of the instructional program. The researchers designed the posttest to be more difficult than the pretest by expanding the range of integers used on the posttest problems to 10 and -10 (as contrasted with 5 to -5 on the pretest) with both one-digit and two-digit correct answers. The following is an example of a subtraction item from the posttest: *Subtract $-7 - 2$* . Cronbach's coefficient alpha for the 24-item posttest was .76.

Attitude questionnaire

A 14-item attitude questionnaire was administered to assess students' attitudes about learning mathematics on the computer and about the amount of instruction received, as well as to assess their satisfaction with the instructional program and their motivation to use similar computer-based instruction in the future. The 14 items used a five-choice Likert-type scale, ranging from strongly agree (4) to strongly disagree (0). The questionnaire was administered in paper-and-pencil form immediately after students completed the posttest. Cronbach's coefficient alpha for the attitude questionnaire was .88.

En route data

An electronic data-tracking system designed as part of the instructional program was used to collect en route data. The system tracked the amount of time each student spent in the program and the options selected under learner control.

Learner control interview protocol

A six-item interview protocol was designed to assess the choices students made under learner control. The questions asked students about the number of times they exercised the control options offered to them, their reasons for doing so, and their general attitudes about having such control.

Procedures

The Learner Preference Scale and the pretest were administered to the participants 3 weeks before the instructional program was delivered. Participants were classified as having high or low preference for control over instruction based on their Learner Preference Scale score. Those with a score above the median score of 23 on the 32-point scale were classified as having a high preference for control, and those with a score below the median were considered as having a low preference for control. The eight students who scored at the median were randomly assigned to high or low preference (four in each group). Participants were blocked by preference for control and randomly assigned within each pair so that one student was in the learner-control version and the other in the program-control version. In total there were 53 students in the high preference-for-control group and 46 in the low preference-for-control group. Fifty students were in the learner-control group and 49 were in the program-control group.

Participants were also categorized as having high and low prior knowledge based on pretest scores. Students above the median pretest score, 7 out of 12, were classified as having high prior knowledge, and those below the median were classified as having low prior knowledge. The 24 students whose scores were at the median were randomly assigned to one of the two groups (13 in high prior knowledge, 11 in low prior knowledge). However, as explained later in the Data Analyses section, prior knowledge was included as a covariate in the form of pretest scored in the primary analysis for achievement and time, rather than as a factor based on the median split. In total there were 50 students categorized as having high prior knowledge and 49 as having low prior knowledge.

The study took place over three 45-min class periods, one per day on consecutive days. Students were informed by the researcher at the beginning of each period that a posttest would be given following their completion of the instructional program and that the posttest would count as part of their regular class grade. Students completed Module 1, Addition, during the first class and Module 2, Subtraction, during the second class. The posttest and the attitude questionnaire were administered during the third period. The researcher interviewed 25 students who participated in a learner-control version of the instruction after the completion of the experimental phase of the study. Students were asked to participate in the interview process on a voluntary basis. Once the pool of volunteers was established, students from within the pool were selected for interviewing by alternating between students with a high preference for control and students with a low preference for control.

Data analyses

One-way analysis of variance was used to analyze the pretest for any existing differences between groups (control type and preference for control), as well as between grade levels and gender. No significant differences were found by control type, preference for control, grade level, or gender.

Pearson Product Moment correlations were calculated to assess the relationship between the pretest scores and each of the following: posttest scores, time in program, optional elements selected under learner control, and attitude scores. Significant relationships were found between pretest and posttest scores, $r = .29$, $p < .01$, and between pretest scores and time in program, $r = -.34$, $p < .01$, but not between pretest scores and optional elements selected, $r = -.01$, or between pretests scores and attitude scores, $r = -.12$. Because of the

significant correlations between pretest scores and posttest scores and between pretest scores and time in program, the general linear model (Maxwell and Delaney 1993; Maxwell et al. 1984) with pretest scores as a quantitative predictor was used to analyze the differences on posttest scores and on time in program among the groups. In other words, prior knowledge was included as a covariate in the form of pretest scores rather than as a factor established through the bivariate median split described in the Procedures section.

The general linear model (GLM) was used in this study instead of analysis of covariance (ANCOVA) because the GLM allows for data analysis with a covariate when the assumptions of ANCOVA fail to be met. ANCOVA is a special case of the GLM that assumes a linear relationship between the covariate and the dependent variable within all levels of the factors. Performing a GLM in which this assumption is met is equivalent to performing an ANCOVA. When this assumption of ANCOVA is not met, the GLM can be tailored to allow more complex analysis that accounts for the violation of this assumption.

In this study, this assumption did not hold true for the achievement variable. A significant three-way interaction for posttest scores was found indicating that the relationship between the posttest and the two factors (preference for control and type of control) varied as a function of pretest scores and therefore was not linear. However, the assumption did hold true for the time-in-program variable since no interaction among time in program, the factors, and the pretest scores was found. Therefore, the GLM performed on the time-in-program variable with pretest scores as a covariate was equivalent to an ANCOVA.

A 2 (preference for control) $\times$ 2 (prior knowledge) ANOVA was used to examine optional elements (examples and practice-and-feedback opportunities) selected within the learner-control version of the program. A 2 (control type) $\times$ 2 (preference for control) $\times$ 2 (prior knowledge) multivariate analysis of variance (MANOVA) was used to analyze differences between groups on the Likert-type attitude items.

Based on the recommendations of Miles and Huberman (1984) and Seidman (1998), interview responses collected from students under the learner-control version of instruction were analyzed by both researchers for common overall themes, as well as themes by groups. The researchers then discussed the nature and number of themes and came to agreement on them before classifying responses under those themes.

Results

The results are reported in this section for achievement, time in program, number of optional elements viewed under learner control, student attitudes, and interviews with participants in the learner control version of instruction.

Achievement

The mean scores and standard deviations for the 24-item posttest are shown by control type (learner and program) and by level of preference for control in Table 2. The overall mean score for students under learner control was 15.88 (66%) and for students under program control was 15.67 (65%), a non-significant difference, $F(1, 91) = 0.74$, $MSE = 14.74$, $p = .39$, $\eta^2 = .01$. The mean scores for preference for control were 15.70 (65%) for the high-preference group and 15.87 (66%) for the low-preference group, also a non-significant difference, $F(1, 91) = 0.51$, $MSE = 14.74$, $p = .48$, $\eta^2 = .01$. Thus, there were no significant main-effect differences for achievement related to either Research Question 1

Table 2 Means and standard deviations of posttest scores by level of preference and control type with pretest scores as a covariate

Preference for control	Learner control	Program control	Totals
High			
<i>M</i>	15.96	15.40	15.70
<i>SD</i>	(4.12)	(4.08)	(4.07)
Low			
<i>M</i>	15.77	15.95	15.87
<i>SD</i>	(3.45)	(4.67)	(3.97)
Total			
<i>M</i>	15.88	15.67	15.78
<i>SD</i>	(3.80)	(4.24)	(4.01)

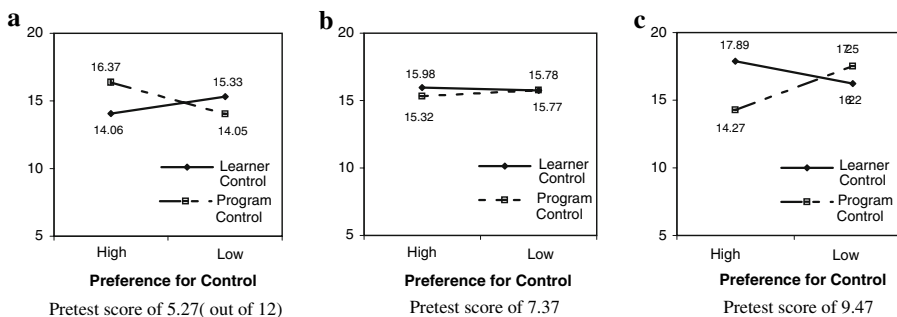
Note. Cell size = 22–28. Posttest scores range from 6 to 24 (out of 24)

(program control versus learner control) or to Research Question 2 (matched versus mismatched preference for control).

Regarding Research Question 3 (interactions for achievement), the general linear model revealed a statistically significant three-way interaction for control type by preference for control by pretest score, $F(1, 91) = 6.36$, $MSE = 14.74$, $p < .05$, $\eta^2 = .07$. A statistically significant two-way interaction for control type by preference for control was also obtained, $F(1, 91) = 5.50$, $MSE = 14.74$, $p < .05$, $\eta^2 = .06$. No other significant interactions were found.

The significant three-way interaction for posttest scores indicates that the relationship between these scores and the two-way interaction between preference for control and type of control varied as a function of pretest scores. To further explore the preference for control by type of control interaction, two new levels of pretest scores were created by subtracting one standard deviation (2.10) from the pretest mean of 7.37 and by adding one standard deviation to the pretest mean. The general linear model was then used to analyze posttest scores by preference for control and type of control at the three different levels of pretest scores: one standard deviation below the mean (5.27), at the mean of 7.37, and at one standard deviation above the mean (9.47).

Figure 2 shows the estimated mean posttest scores under the general linear model for the two-way interaction of preference for control by type of control at each of the three

**Fig. 2** Adjusted mean posttest scores for the three-way interaction between preference for control, type of control, and pretest scores

levels of pretest score. The Fig. 2a reveals that, at the low level of pretest score ($M = 5.27$), students who received the program version that matched their preference for control (students under program control with a low preference for control and those under learner control with a high preference for control) had the lowest estimated mean scores on the posttest, $M = 14.05$ and $M = 14.06$, whereas students who received the version that did not match their preference (received program control but had high preference for control and received learner control but had low preference for control) had the highest estimated mean scores.

In contrast to the low-prior-knowledge students in Fig. 2a and c reveals that at the high level of pretest score (9.47), students who received the program version that matched their preference for control (program control with low preference for control and learner control with high preference) had the highest estimated posttest scores, $M = 17.52$ and $M = 17.89$. Their high prior knowledge counterparts who received the version that was mismatched with their preference for control (program control with high preference and learner control with low preference) had the lowest estimated posttest scores ($M = 14.27$ and $M = 16.22$). The 14.27 estimated posttest mean for the mismatched high-prior-knowledge students with a high preference for control was especially low, more than three points below the estimated means of both high-prior-knowledge groups who received the program version that matched their preference for control. In fact, the estimated pretest mean of this mismatched high-prior-knowledge group, despite their high pretest mean of 9.47 (out of 12), was lower than the estimated means of all four medium-prior-knowledge groups and two of the four low-prior-knowledge groups.

At the mean-prior-knowledge score of 7.37, Fig. 2b shows very little variation in estimated posttest scored for the two levels of preference for control and two types of program. All four estimated posttest means for these groups fell within the narrow range of 15.32–15.98.

Follow-up analyses were conducted to determine the differences on the posttest scores at each of the three levels of the covariate, that is, the pretest. The general linear model revealed a statistically significant two-way interaction between control type and preference for control at the high level of the pretest, $F(1, 91) = 4.50$, $MSE = 66.35$, $p < .05$, $\eta^2 = .05$. The two-way interactions between control type and preference for control at the low level of the pretest and at the mean were non-significant, $F(1, 91) = 2.51$, $MSE = 36.94$, $p = .12$, $\eta^2 = .03$ and $F(1, 91) = 0.19$, $MSE = 2.74$, $p = .67$, $\eta^2 = .01$.

Follow-up analyses were conducted to determine the differences on the posttest scores in the two-way interaction between control type and preference for control at the high level of pretest scores. One-way analysis of variance revealed that high-prior-knowledge students who had a high preference for control and were assigned to learner control ($M = 17.89$) significantly surpassed high-prior-knowledge students who had a high preference for control and were assigned program control ($M = 14.27$), $F(1, 91) = 4.23$, $MSE = 14.74$, $p < .05$, $\eta^2 = .04$. No other statistically significant differences were found.

Time in program

The mean times and standard deviations for time spent in the instructional program are shown by treatment condition in Table 3. The average time by preference for control was 26.55 min by students with a high preference and 20.62 min by students with a low preference. The average time for type of control was 23.91 min by students under program control and 23.67 min for students under learner control.

Table 3 Means and standard deviations for time in program (in minutes) by level of preference and control type with pretest scores as a covariate

Preference for control	Control type		
	Learner	Program	Totals
High			
<i>M</i>	26.28	26.85	26.55
<i>SD</i>	(8.83)	(6.21)	(7.64)
Low			
<i>M</i>	20.91	20.36	20.63
<i>SD</i>	(9.05)	(7.93)	(8.39)
Total			
<i>M</i>	23.92	23.67	23.80
<i>SD</i>	(9.24)	(7.76)	(8.49)

Note. Cell size = 22–28

Because of the significant correlation between pretest scores and time-in-program, the general linear model with pretest scores as a covariate was also used to analyze the differences between groups in time-in-program. The analysis revealed a significant difference for preference for control, $F(1, 94) = 13.59$, $p < .01$, $\eta^2 = .13$, but not for type of control, $F(1, 94) = 0.01$, $p = .97$, $\eta^2 = .01$, nor for the preference for control $\times$ control type interaction. The significant preference for control difference for overall time was further analyzed for difference by type of screens. These analyses revealed that the high-preference subjects spent significantly more time per screen than the low-preference subjects at the $p < .05$ on both example screens and practice screen, but not information screens.

Option use

Research Question 4 dealt with the relationship of students' preference for control and their prior knowledge to their choices of amount of instruction under learner control. Students under learner control could view a total of 40 optional elements, 16 examples and 24 practice-and-feedback choices. Students with a high preference for control selected an average of 36.81 optional elements and those with a low preference selected an average of 31.24. The high-prior-knowledge students chose an average of 34.32 optional elements and the low-prior-knowledge group chose an average of 34.45. The 2 (preference for control) $\times$ 2 (prior knowledge) ANOVA on options viewed yielded a significant difference for preference for control, $F(1, 44) = 4.61$, $p < .05$, $\eta^2 = .15$, but not for prior knowledge, $F(1, 44) = 0.01$, $p = .95$, $\eta^2 = .00$, or for the preference for control by prior knowledge interaction, $F(1, 44) = 0.05$, $p = .82$, $\eta^2 = .01$.

Student attitudes

The mean attitude scores and standard deviations for the 14 Likert-type items on the attitude survey administered after completion of the instructional program are shown by

type of control in Table 4. The mean score on the attitude survey, on which responses were scored from 4 (strongly agree) to 0 (strongly disagree), was 2.79, falling closer to a response of “agree” (3) than “neither agree nor disagree” (2). Four individual items were rated slightly above the agree level of 3.0: “This program taught me a lot about adding and subtracting integers”, “I would like to use computer programs like this one again to learn math”, “I liked the program”, and “I learned a lot from this program”.

A 2 (control type) $\times$ 2 (preference for control) $\times$ 2 (prior knowledge) multivariate analysis of variance revealed a significant overall difference by control type in favor of learner control ($M = 2.89$) over program control ($M = 2.69$), Wilk’s $\Lambda = .75$,

Table 4 Means and standard deviations of attitude scores by item and by control type

#	Item	Control type		<i>p</i>	Total
		LC	PC		
1.	I had some control over the length of the program.	2.79 (0.81)	2.66 (0.87)	.39	2.72 (0.83)
2.	I would like to use computer programs like this one again to learn math.	3.09 (1.10)	3.02 (1.19)	.67	3.05 (1.14)
3.	This program taught me a lot about adding and subtracting integers.	3.13 (0.97)	3.09 (1.12)	.68	3.11 (1.04)
4.**	I had some control over the number of problems I did in the program.	2.70 (0.98)	2.02 (1.22)	.01	2.36 (1.15)
5.*	I was able to adjust the program to get the amount of instruction I wanted.	2.83 (0.84)	2.30 (1.20)	.02	2.56 (1.06)
6.	The number of examples I saw during the program was about right.	2.79 (0.88)	2.96 (0.88)	.29	2.87 (0.88)
7.	I liked the program.	3.02 (1.09)	3.06 (1.07)	.98	3.04 (1.08)
8.	I was satisfied with how I could control the length of the program.	2.77 (0.91)	2.45 (0.97)	.07	2.61 (0.95)
9.	I want to learn more about adding and subtracting integers.	2.83 (1.09)	2.68 (1.07)	.36	2.76 (1.07)
10.	I learned a lot from this program.	3.09 (1.00)	2.96 (1.12)	.52	3.02 (1.06)
11.**	I could control the number of examples I saw during the program.	2.72 (0.97)	2.11 (1.11)	.01	2.41 (1.08)
12.	I would recommend that other students use this program.	2.89 (1.15)	3.02 (0.82)	.50	2.96 (0.99)
13.	I had about the right number of problems to do in order to learn how to add and subtract integers.	2.77 (1.03)	2.72 (0.93)	.93	2.74 (0.95)
14.*	I knew when I could skip over part of the instruction.	3.06 (0.92)	2.62 (1.07)	.05	2.84 (1.02)
	*Totals	2.89 (0.62)	2.69 (0.64)	.02	2.79 (0.63)

Note. $N = 94$. Items rated on a five-point scale, from strongly disagree (0) to strongly agree (4)

Items marked with a double asterisk (**) are significant at $p < .01$

Items marked with a single asterisk (*) are significant at $p < .05$

$F(14, 73) = 2.14, p < .05, \eta^2 = .29$. Because no significant main effect differences were found for preference for control or prior knowledge, the mean attitude scores for these factors are not presented in Table 4.

Follow-up univariate analyses of variance were conducted to analyze the differences in ratings for each item by control type. Six of the 14 items on the attitude survey (items 1, 4, 5, 8, 11, and 14) dealt with the topic of control over the various elements of instruction, and significant differences in favor of learner control were obtained on four of these items. These items were “I had some control over the number of problems I did in the program”, “I knew when I could skip over part of the instruction”, “I could control the number of examples I saw during the program”, and “I was able to adjust the program to get the amount of instruction I wanted”. These four items were the only ones on which significant differences between learner control and program control were obtained.

The MANOVA also revealed a significant three-way interaction between control type, preference for control, and prior knowledge, Wilk's $\Lambda = .71, F(14, 73) = 2.12, p < .05, \eta^2 = .29$. Univariate analyses of variance were conducted to analyze the differences in ratings by individual item. Significant three-way interactions were found on three of the 14 items: “This program taught me a lot about adding and subtracting integers”, “I liked the program”, and “I want to learn more about adding and subtracting integers”.

For each of these three items, the interaction between attitudes, preference for control, and control type changes as a function of prior knowledge, just as it did for posttest scores. The significant three-way interaction can be seen by examining the mean scores for each group on these three items as shown in Table 5. At a low level of prior knowledge, high-preference-for-control students had higher agreement when their preference for control was mismatched with the version of instruction they received (that is, they preferred more learner control but received program control) than when their preference for control matched the version of instruction they received (they preferred learner control and received learner control). Similarly, low-prior-knowledge students who had a low preference for control had higher agreement with all three items when they received a version of instruction that did not match their preference (learner control) than when they received a version that matched their preference (program control).

Table 5 Means and standard deviations of attitude scores by level of preference, control type, and prior knowledge on items with a significant three-way interaction

Item	Preference for control	Learner control		Program control	
		High prior knowledge	Low prior knowledge	High prior knowledge	Low prior knowledge
**3. This program taught me a lot about adding and subtracting integers.	High	3.36 (1.08)	2.73 (0.90)	2.92 (1.00)	3.67 (1.29)
	Low	2.83 (0.94)	3.60 (0.70)	3.14 (0.49)	2.44 (1.33)
**7. I liked the program.	High	3.36 (1.08)	2.64 (1.29)	2.83 (1.19)	3.50 (0.52)
	Low	2.75 (1.14)	3.30 (0.67)	3.29 (0.83)	2.44 (1.51)
*9. I want to learn more about adding and subtracting integers.	High	3.00 (0.88)	2.73 (1.42)	2.75 (0.87)	3.17 (0.39)
	Low	2.50 (1.17)	3.10 (0.88)	2.71 (1.07)	1.89 (1.54)
Items 3, 7, and 9 combined for follow-up comparisons.	High	3.24 (1.01)	2.7 (1.20)	2.83 (1.02)	3.45 (0.73)
	Low	2.69 (1.08)	3.33 (0.75)	3.05 (0.80)	2.26 (1.46)

Note. The three-way interaction for items marked with a double asterisk (**) is significant at $p < .01$

The three-way interaction for the item marked with a single asterisk (*) is significant at $p < .05$

This pattern reverses at a high level of prior knowledge. High-prior-knowledge students with a high preference for control had higher agreement across all three items when they received the version of the program that matched their preference (learner control) and the least agreement when they received the version of the program that was mismatched (program control). High-prior knowledge students with a low preference for control also had higher agreement across all three items when they received the version of instruction that matched their preference (program control) than when the version did not match their preference (learner control).

To further explore the specific nature of the three-way interaction on these three items, the attitude scores for the three items were combined into one mean score (as shown in Table 5) to avoid the redundancy of explaining the same interaction three separate times. Post hoc testing was conducted on the combined mean scores in order to determine the differences in attitude scores of high and low preference-for-control subjects. The alpha level for these follow-up tests was set at .01 because of the number of cells (eight) in the pairwise comparisons. The highest mean scores for the eight cells were 3.45 and 3.33 for the two low-prior-knowledge groups whose treatments were mismatched with their preference for control and 3.24 and 3.05 for the two high-prior-knowledge groups whose treatment matched their preferences for control. The lowest mean score was 2.26 for the low-prior-knowledge subjects whose program-control treatment matched their preference for control (low). The pairwise comparisons of each of the three highest mean scores with the lowest mean were significant at the .01 level (the fourth comparison yielded a *p*-value of .03). There were no other significant differences in attitudes at the .01 level in any of the follow-up comparisons.

Interviews

A total of 25 students (11 with a high preference for control and 14 with a low preference for control) who received the learner-control version of the instruction participated in individual follow-up interviews regarding the choices they made under learner control. In response to the question “Did you use the SKIP button during the program?” 19 of the 25 students said “Yes”. The most frequently stated reason for using it was because the students knew the material already (9 students). Of 6 who did not use the SKIP button, 5 had scored high on the preference for control scale, stating that they did not use it because they wanted to be sure they understood all of the instruction. In general, students with a high preference for control and those with a low preference responded similarly to the other interview questions.

A consistent theme that emerged from the interview responses was that if students already knew the material, they should be given control that allowed them to skip parts of the instruction. Of the 25 students interviewed, 12 stated it was good to have the SKIP button in the program because if the instruction was something they already understood, they should be able to skip it. Also, 13 felt that having the SKIP button made learning easier. When asked if every student should be able to skip parts of the instruction, 18 responded “yes” and 6 further explained that if students already knew the material, they should be able to skip it.

Discussion

The present study was designed to examine the effects of learner control and program control of computer-based mathematics instruction on the achievement, attitudes, and

time-in-program of middle-school students with differing levels of prior knowledge and preference for control. The results are discussed below for achievement, attitudes, time-in-program, and option use within the learner-control condition.

Achievement

The significant three-way interaction (preference for control by type of control by level of prior knowledge in the form of pretest scores) in this study reflected a pattern indicating that students with high prior knowledge scored higher on the posttest when they were assigned to the version of the instructional program that matched their preference for control, whereas students with low prior knowledge scored higher when their preference for control was mismatched with the version of instruction they received. The general implication of these results is that giving learners control over the amount of instruction they receive may be an effective strategy for those with a high level of prior knowledge or ability, but not for those with a low level. Several researchers (Carrier 1984; Hannafin 1984; Williams 1996) have reported that learners with higher ability or higher levels of prior knowledge are more aware of their preferences for learning and therefore are able to make better decisions about how to learn best.

Lower-ability students, on the other hand, may be less aware of their preferences or may be more likely to prefer a version of instruction that enables them to finish more quickly or more easily, rather than one that may help them achieve more. Farrell and Moore (2000–2001) and Gay (1986) found that giving lower-ability students control over aspects of instruction does not yield better achievement than program control.

The statistically significant difference between the very low estimated mean score ($M = 14.27$) for high-prior-knowledge students who had a high preference for control, but were assigned to program control, and the estimated mean score ($M = 17.89$) for their counterparts who were assigned learner control was particularly interesting. This result seems most likely to be due to the high-prior-knowledge subjects being in a treatment condition that did not match their preference for control and/or that required them to receive more instruction than they thought they needed. It appears to be an instance of the expertise reversal effect described by Kalyuga et al. (2003). Reisslein and her colleagues have encountered this effect in two recent engineering-related studies involving high-prior-knowledge students who scored much lower on the posttest under conditions in which they received more instruction and practice than under conditions in which they received less instruction and practice (Reisslein et al. 2006, 2007).

Option use

Although there was only a small non-significant main-effect difference in posttest achievement by level of preference for control, there was a significant difference in option use between students with a high preference for control (36.81 of 40 options used) and those with a low preference (31.24 options used). This finding suggests that the Learner Preferences Scale developed by the researchers as a measure of preference for control for this study may also be somewhat accurate as an indicator of the amount of instruction that learners will select under learner control. Its usefulness for this purpose is also substantiated to a large degree by the fact that 5 of the 11 learner-control students with a high

preference for control reported in their interviews that they did not use the skip button at all, whereas only 1 of the 14 low-preference students reported not using it.

Learner control and learner preferences are important concepts in much of the recent writing by cognitive learning theorists (Bannert 2002; Corbalan et al. 2006; van Merriënboer et al. 2002; van Gog et al. 2005). The Learning Preference Scale in this study did differentiate between high-preference and low-preference for control subjects both in amount of option use and time-in-program. This fact, combined with its relatively high reliability ($r = .82$) for a short scale, suggests that adaptations of it could be useful in other studies of learner preferences for control or for amount of instruction. The scale for this study was developed specifically for mathematics instruction. It could easily be modified for instruction in other subject areas or for instruction in general just by changing the name of the subject area on the scale or by adding more subject areas and possibly reducing the number of items in each area.

Time in program

Subjects with a high preference for control not only selected more optional elements under learner control, but also spent significantly more time in the instructional program (26.55 min for high preference, 20.62 min for low preference). Their pattern of greater time in program was consistent across both the learner-control treatment, in which the high-preference subjects viewed significantly more optional elements than subjects with a low preference for control, and across the program-control treatment, in which both high-preference and low-preference subjects viewed all 40 optional elements and all 249 screens. In fact, the difference in time-in-program between subjects with a high preference for control and those with a low preference for control was slightly greater under program control (approximately 6 min more for high-preference subjects) than for those under learner control (approximately 5 min more for high-preference subjects). Thus, the greater amount of time-in-program for subjects with a high preference for control was due not only to the fact these subjects chose more optional screens under learner control, but to the fact that they spent significantly more time on the screens that they did view.

Student attitudes

Students under learner control had significantly more positive responses to the attitude survey than those under program control, due almost entirely to their responses on the six questions about the amount of control they could exercise. Subjects under learner control responded more favorably than those under program control on all six of these items, and significantly more positively on four of the items. The four items on which significant differences occurred dealt with awareness that they had control over the number of problems and examples and with the knowledge that they could adjust the program and skip over part of the instruction. Thus, these items were more knowledge items than preference items.

Some researchers have noted that subjects under learner control may not know that they have that control (Farrell and Moore 2000–2001; Kinzie 1990). That students under learner control responded more positively to the items dealing with overall control in this study indicates quite clearly that they were aware of the control they could exercise under the learner-control condition. However, there was no significant evidence from the attitude

survey that subjects generally liked one version of the program better than the other or learned more from either version.

Like the achievement data, the relationship between attitude scores, preference for control, and control type also changed as a function of prior knowledge on three items: “This program taught me a lot about adding and subtracting integers”, “I liked the program”, and “I want to learn more about adding and subtracting integers”. High-prior-knowledge students had more positive attitudes on these items when they received the version of instruction that matched their preference for control and low-prior-knowledge students had more positive attitudes when they received the version of instruction that did not match their preference for control. The similarity between achievement scores and attitude scores on these three items indicates that high-preference and low-preference students were accurate in their judgments about how much the program “taught them”, that they liked the version better from which they learned more, and that they had greater continuing motivation associated with the version they learned more from and liked better. This result is especially interesting for low-prior-knowledge students because it indicates that, despite their initial preference for control, they preferred the version from which they learned more upon completing the program.

Summary and conclusion

Several findings from this study are summarized below.

1. Research Questions 1 and 2: Neither the variation in preference for amount of student control nor the type of control (learner or program) resulted in a significant main-effect difference in achievement.
2. Research Question 3: The significant three-way interaction for achievement (prior knowledge by preference for control by type of control) indicates that high-prior-knowledge students achieved better when their preference for control was matched with the type of program they received and that low-prior-knowledge students achieved somewhat better when their preference for control and type of control were mismatched. The achievement pattern for high-prior-knowledge students provides evidence supporting the expertise reversal effect described by Kalyuga et al. (2003).
3. The three-way interaction for attitudes paralleled the three-way interaction for achievement. Nevertheless, low-prior-knowledge students’ end-of-program attitude scores indicated that they preferred the version from which they learned more, suggesting that what low-prior-knowledge students prefer initially may not be consistent with their post-program attitudes.
4. Research Question 4: There were significant relationships between student scores on the Learner Preference Scale and both number of options used in the program and time spent in the program. The authors suggest that adaptations of the scale may be appropriate for use in future learner-control studies.

A number of researchers have suggested that instructional environments should allow students to have control over the learning process (Farrell and Moore 2000–2001; Kinzie 1990; van Gog et al. 2005; van Merriënboer et al. 2002). The results of this study lend support to researchers (Merrill 2002; Corbalan et al. 2006) who recommend designing learner-controlled environments so that more control is provided to learners as the learners’ level of knowledge increases. Designing web-based or distance learning environments to monitor the learners’ knowledge level and adjust the provision of instructional control

accordingly has the potential to improve learner achievement and attitudes about the instruction they receive.

Under learner-control conditions, students with lower ability or lower prior knowledge may not receive enough instruction to enable them to achieve well. In general, students in a learner-controlled program often fail to add instruction when they need it (Eom and Reiser 2000; Hannafin 1984; Morrison et al. 1992). The basic instructional program in this study was developed by the researchers to be more of a full version rather than a lean one, based on the recommendations of Hannafin and Sullivan (1995) and Schnackenberg et al. (1998). The use of a full version appears to have been appropriate. Learners in this study skipped a relatively small amount of optional instruction under learner control (8% for high-preference-for-control students and 22% for low-preference-for-control students), and overall they achieved at a similar level under learner and program control. Designers of learner-controlled instruction should consider making the basic instructional program in learner-controlled environments one that is more full than lean. Students under learner control, especially those with low prior knowledge, are likely to receive much less instruction in a lean program than in a full one.

Learners with higher amounts of prior knowledge are more likely to have the skills and cognitive resources needed to make decisions about instruction that will lead to improved learning (Merrill 2002; van Gog et al. 2005). This was true in the present study. Learners with higher levels of prior knowledge were able to indicate their learning preferences and exercise control when they preferred it in a way that helped them achieve better. This is promising for researchers interested in reducing the cognitive load of learners in online instruction by affording them control over that instruction. Measuring the preference for control of high-prior-knowledge learners and assigning control according to their stated preference is likely to be an effective approach to reducing their cognitive load during instruction.

References

- Alomyan, H. (2004). Individual differences: Implications for web-based learning design. *International Education Journal*, 4(4), 188–196.
- Bannert, M. (2002). Managing cognitive load – recent trends in cognitive load theory. *Learning and Instruction*, 12, 139–146.
- Carrier, C. A. (1984). Do learners make good choices? A review of research on learner control in instruction. *Instructional Innovator*, 29(2), 15–17.
- Carrier, C. A., Davidson, G., Higson, V., & Williams, M. (1984). Selections of options by field independent and dependent children in a computer-based concept lesson. *Journal of Computer-Based Instruction*, 11(2), 49–54.
- Carrier, C. A., Davidson, G., & Williams, M. (1985). Selection of instructional options in a computer-based coordinate concept lesson. *Educational Communications and Technology Journal*, 33(3), 199–212.
- Chen, S. Y., & Macredie, R. D. (2002). Cognitive styles and hypermedia navigation: Development of a learning model. *Journal of the American Society for Information Science and Technology*, 53(1), 3–15.
- Corbalan, G., Kester, L., & van Merriënboer, J. J. G. (2006). Towards a personalized task selection model with shared instructional control. *Instructional Science*, 34, 399–422.
- Eom, W.-Y., & Reiser, R. A. (2000). The effects of self-regulation and instructional control on performance and motivation in computer-based instruction. *International Journal of Instructional Media*, 27(3), 247–260.
- El-Tigi, M., & Branch, R. M. (1997). Designing for interaction, learner control, and feedback during web-based learning. *Educational Technology*, 37(3), 23–29.
- Farrell, I. H., & Moore, D. M. (2000–2001). The effect of navigation tools on learners' achievement and attitude in a hypermedia environment. *Journal of Educational Technology Systems*, 29(2), 169–181.
- Freitag, E. T., & Sullivan, H. J. (1995). Matching learner preference to amount of instruction: An alternative form of learner control. *Educational Technology Research and Development*, 43(2), 5–14.

- Gay, G. (1986). Interaction of learner control and prior understanding in computer-assisted video instruction. *Journal of Educational Psychology*, 78, 225–227.
- Gerjets, P., & Scheiter, K. (2003). Goal configurations and processing strategies as moderators between instructional design and cognitive load: Evidence from hypertext-based instruction. *Educational Psychologist*, 38, 33–41.
- Gray, S. H. (1987). The effect of sequence control on computer assisted learning. *Journal of Computer-Based Instruction*, 14, 54–56.
- Hannafin, M. J. (1984). Guidelines for using locus of instructional control in the design of computer-assisted instruction. *Journal of Instructional Development*, 7(3), 6–10.
- Hannafin, R. D., & Sullivan, H. J. (1995). Learner control in full and lean CAI programs. *Educational Technology Research and Development*, 43(1), 19–30.
- Hannafin, R. D., & Sullivan, H. J. (1996). Learner preferences and learner control over amount of instruction. *Journal of Educational Psychology*, 88(1), 162–173.
- Hicken, S., Sullivan, H., & Klein, J. D. (1992). Learner control modes and incentive variations in computer-delivered instruction. *Educational Technology Research and Development*, 40(4), 15–26.
- Kalyuga, S., Ayers, P., Chandler, P., & Sweller, J. (2003). The expertise reversal effect. *Educational Psychologist*, 38(1), 23–31.
- Kinzie, M. B. (1990). Requirements and benefits of effective interactive instruction: Learner control, self-regulation, and continuing motivation. *Educational Technology Research and Development*, 38(1), 1–21.
- Kinzie, M. B., Sullivan, H. J., & Berdel, R. L. (1988). Learner control and achievement in science computer-assisted instruction. *Journal of Educational Psychology*, 80(3), 299–303.
- Klein, J. D., & Keller, J. M. (1990). Influence of student ability, locus of control, and type of instructional control on performance and confidence. *Journal of Educational Research*, 83(3), 140–146.
- Lawless, K. A., & Brown, S. W. (1997). Multimedia learning environments: Issues of learner control and navigation. *Instructional Science*, 25(2), 117–131.
- Maxwell, S. E., & Delaney, H. D. (1993). Bivariate median splits and spurious statistical significance. *Psychological Bulletin*, 113(1), 181–190.
- Maxwell, S. E., Delaney, H. D., & Dill, C. A. (1984). Another look at ANCOVA versus blocking. *Psychological Bulletin*, 95(1), 136–147.
- Merrill, M. D. (1975). Learner control: Beyond aptitude–treatment interactions. *AV Communications Review*, 23, 217–226.
- Merrill, M. D. (1980). Learner control in computer based learning. *Computers and Education*, 4, 77–95.
- Merrill, M. D. (1984). What is learner control? In R. K. Bass & C. D. Dills (Eds.), *Instructional development: The state of the art II*. Dubuque, IA: Kendall/Hunt (ERIC Document Reproduction Service No. ED298 905).
- Merrill, M. D. (2002). Instructional strategies and learning styles: Which takes precedence? In R. A. Reiser, & J. V. Dempsey (Eds.), *Trends and issues in instructional technology*. Upper Saddle River NJ: Merrill Prentice Hall.
- Miles, M. B., & Huberman, A. M. (1984). *Qualitative data analysis: A sourcebook of new methods*. Newbury Park, CA: Sage.
- Morrison, G. R., Ross, S. M., & Baldwin, W. (1992). Learner control of context and instructional support in learning elementary school mathematics. *Educational Technology Research & Development*, 40(1), 5–13.
- Reisslein, J., Atkinson, R. K., Seeling, P., & Reisslein, M. (2006). Encountering the expertise reversal effect with computer-based learning environment in electrical circuit analysis. *Learning and Instruction*, 16(2), 92–103.
- Reisslein, J., Sullivan, H., & Reisslein, M. (2007). Learner achievement and attitudes under different paces of transitioning to independent problem solving. *Journal of Engineering Education*, 96(1), 45–55.
- Salden, R. J. C. M., Paas, F., & van Merriënboer, J. J. G. (2006). Personalised adaptive task selection in air traffic control: Effects on training efficiency and transfer. *Learning and Instruction*, 16, 350–362.
- Schmidt, A. M., & Ford, J. K. (2003). Learning within a learner control training environment: The interactive effects of goal orientation and metacognitive instruction on learning outcomes. *Personnel Psychology*, 56, 405–429.
- Schnackenberg, H., & Sullivan, H. J. (2000). Learner control over full and lean computer-based instruction under differing ability levels. *Educational Technology Research and Development*, 48(2), 19–35.
- Schnackenberg, H., Sullivan, H. J., Leader, L. F., & Jones, E. E. K. (1998). Learner preference and achievement under differing amounts of learner practice. *Educational Technology Research and Development*, 46(2), 5–15.
- Seidman, I. (1998). *Interviewing as qualitative research: A guide for researchers in education and the social sciences*. New York: Teachers College Press.

- Shin, E. J., Schallert, D., & Savenye, W. C. (1994). Effects of learner control, advisement & prior knowledge on young students' learning in a hypertext environment. *Educational Technology Research and Development*, 42(1), 33–46.
- Shyu, H.-Y., & Brown, S. W. (1992). Learner control versus program control in interactive videodisc instruction: What are the effects in procedural learning? *International Journal of Instructional Media*, 19(2), 85–96.
- Swaak, J., & de Jong, T. (2001). Learner vs. system control in using online support for simulation-based discovery learning. *Learning Environments Research*, 4, 217–241.
- Triantafillou, E., Pomportsis, A., & Demetriadis, S. (2003). The design and the formative evaluation of an adaptive educational system based on cognitive styles. *Computers and Education*, 41, 87–103.
- van Gog, T., Ericsson, K. A., Rikers, R. M. J. P., & Paas, F. (2005). Instructional design for advanced learners: Establishing connections between the theoretical frameworks of cognitive load and deliberate practice. *Educational Technology Research and Design*, 53(3), 73–81.
- van Merriënboer, J. J. G., Schuurman, J. G., de Croock, M. B. M., & Paas, F. G. W. C. (2002). Redirecting learners' attention during training: Effects on cognitive load, transfer test performance and training efficiency. *Learning and Instruction*, 12, 11–37.
- Wang, L.-C. C., & Sutton, R. E. (2002). Effects of learner control with advisement on English-as-a-Second-Language students in a web-based learning environment. *International Journal of Instructional Media*, 29(3), 317–324.
- Williams, M. D. (1996). Learner-control and instructional technologies. In D. Jonassen (Ed.), *Handbook of research for educational communications and technology* (2nd ed.). Mahwah, NJ: Lawrence Erlbaum Associates.
- Yang, Y.-C., & Chin, W.-K. (1996). Motivational analyses on the effects of type of instructional control on learning from computer-based instruction. *Journal of Educational Technology Systems*, 25(1), 25–35.

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